

- 4 (a) The transformation T is represented by the matrix $\mathbf{M} = \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & 0 \\ 1 & 2 & -1 \end{pmatrix}$.

A shape S_1 is mapped to a shape S_2 by the transformation T .

Show that volume of S_1 is the same as the volume of S_2 . [2]

- (b) Three planes have equations

$$\begin{aligned}x - 2y + 2z &= \lambda, \\ 2x + y &= 2, \\ x + 2y - z &= 0,\end{aligned}$$

where λ is a constant.

- (i) Explain why the three planes intersect at a point for any value of λ . [2]

- (ii) Use a matrix method to determine, in terms of λ , the coordinates of this point. [4]